

Inhibitory effect of arctigenin on reovirus infection in grass carp

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ABSTRACT

Chinese herbal medicine can effectively prevent and treat viral diseases in China. It not only affects the replication of the virus but also regulates the immune function of the body. *Arctium lappa* L. (*A. lappa*) monomer Arctigenin (AR) has anti-inflammatory, immunomodulatory, antiviral, antitumor, and antibacterial activities *in vivo* and *in vitro*. Several studies have been conducted on mammals and aquatic animals, but the effect of AR on grass carp reovirus (GCRV) is rarely reported. To develop effective drugs for controlling GCRV infection, the Chinese herbal active monomer AR was chosen to evaluate its anti-GCRV activity. First, the maximum safe concentration of AR on CIK cells was determined by the CCK-8 assay. Then, the inhibitory effect of different concentrations of AR on the GCRV was quantitatively determined by fluorescence. The results showed that the antiviral activity of AR was dose-dependent. CIK cells were treated with 80 µg/mL AR in three different ways. AR significantly inhibited the proliferation of GCRV. The results of the add time analysis showed that AR can reduce the infectivity of the virus and inhibit the invasion of GCRV. AR mainly inhibited the early replication of GCRV but did not interfere with its adsorption. The images obtained by fluorescence inverted microscope showed that AR could protect the integrity of CIK cells from GCRV-induced nuclear fragmentation, cytoplasmic degradation, and apoptosis. The results from flow cytometry, fluorescence quantification and Western blotting showed that AR inhibits GCRV replication by blocking GCRV-induced apoptosis *in vitro*. AR also showed strong antiviral activity *in vivo* and inhibited the expression of the *S4* gene of GCRV. Histopathological evaluation showed pathological changes in the liver and intestinal tissues of grass carp infected with GCRV, but no tissues were damaged in either the normal or the AR-treated groups. *In vivo*, AR can also effectively regulate related apoptosis genes to achieve antiviral effects and downregulate the expression of cytokines (IL-1β, IL-6 and TNF-α). To summarize, AR can be used as a potential drug for treating GCRV infection. This study laid the foundation for developing traditional Chinese medicine monomers in aquaculture.

1. Introduction

Traditional Chinese medicine has many functions, for example, it can be used as a feed additive, facilitate disease prevention, improve healthcare, and promote production performance (Gong et al., 2021; Liu et al., 2023; Song et al., 2023; Zhao et al., 2022). It can also be used as an immune adjuvant, promote the development of immune organs, enhance the activity of immune cells, and regulate the immune function of the body (Feng et al., 2024; Li et al., 2021a; Liu et al., 2021; Liu et al., 2019; Zhao et al., 2023). *Arctium lappa* L. (*A. lappa*) is a dried and ripe fruit of the biennial herb burdock from the family *Asteraceae*. It can clear heat and detoxify, clear the lungs, relieve asthma, resolve muscle, reduce rashes, etc. It is mainly used to treat external wind and heat,

throat swelling, fever, cough, measles, etc. (Chan et al., 2011; Yosri et al., 2023). A study showed that the root extract of *A. lappa* had a protective effect on the liver of model mice (Lin et al., 2002). In *Arctium lappa* L., the physiological and pharmacological effects are mainly exerted by lignans, such as Arctiin and Arctigenin (AR) (Gao et al., 2018). Several studies have shown that AR has antiviral, anti-inflammatory, insecticidal, anti-tumor, and immunomodulatory activities (Hayashi et al., 2010; Lee et al., 2011; Zhang et al., 2018a; Zhao et al., 2020). For example, AR decreases excessive microglial activation and prevents neuroinflammation by influencing the HMGB1/TLR4/NF-κB and TNF-α/TNFR1/NF-κB signaling pathways, thus exhibiting antidepressant effects (Xu et al., 2020). AR can alleviate immune dysfunction and counteract the over-activation of keratinocytes in the inflamed

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skin microenvironment, thus, it may be a potential treatment for psoriasis (Li et al., 2021b). AR can inhibit the growth of liver cancer by directly recruiting C/EBP α to gankyrin promoters (Sun et al., 2018). AR can strongly inhibit HIV-1 by acting on the assembly stage of the virus. It can also significantly inhibit the expression of P17 and P24 proteins of the HIV-1 virus *in vitro*, suggesting that AR may also act on the reverse transcription stage (Eich et al., 1996; Vlietinck et al., 1998). AR can also significantly inhibit the proliferation of porcine circovirus (PCV2) in the lung, spleen, and inguinal lymph nodes. Such multifaceted effects of AR can contribute to the advancement of Chinese herbal medicine for the treatment of PCV2-related diseases (Chen et al., 2016). AR can also be administered for treating fish viruses. Through the activation of the JNK pathway at the early stage of viral replication, *epithelioma papulosum cyprini* (EPC) cells were induced to perform autophagy, which eventually inhibited the replication of infectious hematopoietic necrosis virus (IHNV) and virion release (Hu et al., 2019a). Most drugs are still developed from the natural products extracted from plants. Their application can reduce the overuse of antibiotics and prevent the development of drug resistance in viruses. Therefore, drug monomers and derivatives isolated from plants have wide applicability in the field of medicine.

Grass carp (*Ctenopharyngodon idella*) is one of the four main fish species in China and an important economic aquaculture fish. Grass carp reovirus (GCRV, double-stranded RNA virus) is the first fish virus isolated in China, belonging to the family *Reoviridae* (aquatic *Reoviridae*) (Wang et al., 2012). It causes severe bleeding and mass death of grass fry. GCRV greatly hinders the development of the aquaculture industry. The known GCRV strains can be divided into three groups, including GCRV-I (representing strain GCRV873), GCRV-II (representing strain GCRVHZ08), and GCRV-III (representing strain GCRV104) (Pei et al., 2014). The strain GCRV873 was isolated from grass carp infected with epidemic hemorrhagic disease in Shaoyang, Hunan, China in 1987 by the Wuhan Institute of Virology, Chinese Academy of Sciences. This strain is considered to be the original strain or standard strain of GCRV by researchers around the world (Ke et al., 2011; Nibert and Duncan, 2013). Many researchers have used GCRV and CIK cells as infection models to study viral structure, infection mechanisms, antiviral drug screening, and vaccine development *in vitro* (Reshi et al., 2016; Tian et al., 2013). In aquaculture, natural products, derivatives, and chemical agents can be used to inhibit viral infection (Zhao et al., 2011; Zhi et al., 2011). Many studies on GCRV have assessed ways to use traditional Chinese medicine for treatment. For example, ginsenoside Rg3 promotes the proliferation of grass carp ovarian (CO) epithelial cells, inhibits the activity of GCRV, and enhances the resistance of CO cells to GCRV infection by promoting the immune activity of CO cells (Dai et al., 2020). Honokiol promotes the expression of innate immune-related genes to enhance innate immune signaling against GCRV infection (Chen et al., 2017). However, studies on the antiviral effects of AR on GCRV are lacking. The cell culture characteristics, virulence, pathogenesis, and antigenicity of each strain may differ considerably due to the differences in the sequences between the strains. In this study, the antiviral effects of AR on GCRV were evaluated and elucidated its underlying mechanism of action *in vitro* and *in vivo*. We also assessed the antiviral effect of AR on CIK cells infected with GCRV-I (strain GCRV873), the protective effect on CIK cells, and the effects on the innate immune response of the host. Our results showed that AR had antiviral effects on GCRV in aquaculture.

2. Materials and methods

2.1. Cells, viruses, antibodies, grass carp culture, and antiviral drugs

The CIK cell line was donated by associate Professor Jie Zhang (Institute of Hydrobiology of the Chinese Academy of Sciences, Wuhan, Hubei Province). All cells were grown at 28 °C in a medium (MEM, Invitrogen, USA) containing 10% fetal bovine serum (Gibco BRL,

Rockville, MD, USA), 100 μ g/mL streptomycin, and 100 μ g/mL penicillin (Gibco, USA) (Tan et al., 2023). GCRV873 (Type I strain) was donated by Professor Qin Fang (Wuhan Institute of Virology of the Chinese Academy of Sciences, Wuhan, Hubei Province), where it was grown in CIK cells at 28 °C and used for infection and reproduction. The GCRV polyclonal antibodies used in this study were provided by the animal important pathogen biology laboratory, Yangtze University. Other antibodies were purchased from BIODOG. The alkaline phosphate-coupled goat anti-mouse IgG and goat anti-rabbit IgG were purchased from Sigma-Aldrich (Burlington, MA, USA). Grass carp (2.5 $\pm$ 0.5 g) were purchased from Hubei Province and cultured in a circulating tank at 28 $\pm$ 1 °C. They were fed commercial feed twice a day, depending on the weight of the fish. AR (HPLC $\geq$ 98%) was purchased from Yuanye (Shanghai, China). It was dissolved in dimethyl sulfoxide (DMSO) (Sigma) before use.

2.2. Cytotoxicity assays

To conduct the cytotoxicity assay, CIK cells were inoculated in 96-well plates containing 100 μ L of growth medium. The cells were plated at a density of 1×10^5 cells per well and incubated until the cell volume reached 80–90%. The cell maintenance medium (MEM medium with 2% FBS) containing different concentrations of AR was then added and incubated for 24 h; untreated cells served as the control for the experiment (Wang et al., 2021). After incubation, cell viability was assessed using a cell counting kit (CCK-8, Biosharp, China), following the manufacturer's instructions. The activity of cells was determined by measuring their absorbance at 450 nm using a microplate reader (Molecular Devices, China). Cell viability was calculated using a formula, and a cell survival rate of 90% was considered to be the safest concentration of the drug. Finally, a certain concentration of the drug was selected for subsequent antiviral assays.

2.3. Virus titration reduction assays

Three different modes of infection were designed to detect the inhibitory effect of AR on GCRV. Viral reproduction and titration were determined according to the assay (Yu et al., 2014). First, CIK cells were cultured into monolayers in 96-well plates. Then, AR was diluted to 80 μ g/mL with MEM medium and co-incubated with GCRV (1×10^3 50% tissue culture infection dose (TCID₅₀)) at 28 °C for 2 h. The drug-virus mixture was added to monolayer cells without the medium and incubated at 28 °C for 2 h. The cells were washed thrice after the culture medium was imbibed, and then, MEM cell maintenance solution containing 2% FBS was added and cultured at 28 °C for 24 h. In the second method, AR was diluted with the MEM medium (80 μ g/mL) and added to the monolayer cells without the medium, followed by incubation at 28 °C for 2 h. After the cells absorbed the drug, a serum-free medium containing GCRV venom was added, followed by incubation for 2 h for the cells to absorb the venom. Next, the cells were washed thrice, and then, the MEM cell maintenance solution containing 2% FBS was added. Finally, the cells were cultured at 28 °C for 24 h. In the third method, the CIK cells were initially infected with GCRV venom for 2 h at 28 °C. Then, the venom was removed, the cells were washed thrice, and the cell maintenance solution (2% FBS MEM) was added. This was followed by diluting to 80 μ g/mL to form a drug maintenance solution, which was incubated for 24 h (Zhou et al., 2022). The untreated cells were used as blank controls, and the positive control cells were administered the same concentration of DMSO (0.8%) as the drug-treated cells. After the cells treated using the above three methods were incubated for 24 h. Each sample was directly observed and photographed under an inverted microscope (SOPTOP ICX41, China).

2.4. Plaque test

CIK cells were inoculated in the six-well plates at a density of $2.5 \times$

10^6 cells per well. All cells were treated with the viruses and AR and cultured at 28 °C for 24 h. All samples were stained with crystal violet staining and visualized under an inverted microscope (SOPTOP ICX41, China).

2.5. RNA isolation cDNA synthesis and detection of mRNA expression by qPCR

Total mRNA of each sample was extracted using Trizol (Biosharp, China), following the manufacturer's protocol. The extracted RNA was reverse transcribed using HiScript II Q Select RT SuperMix for qPCR (+gDNA wiper) (Vazyme, China) (Chu et al., 2022). The CFX96 Real-Time System (Bio-Rad, USA) was used for quantitative PCR assays using the ChamQ Universal SYBR qPCR Master Mix (Vazyme, China) with the following parameters: 95 °C for 30 s, followed by 40 cycles (10 s at 95 °C and 30 s at 60 °C). The relative copy number of GCRV was calculated by the $2^{-\Delta\Delta Ct}$ method. Primer sequences designed with reference to previous studies (Supplementary Table 1).

2.6. Western blotting analysis

The drug and virus treated the cells for 24 h, they were harvested by washing thrice with PBS, and then, they were lysed on ice. Next, the cells were centrifuged, boiled with a protein loading buffer, and transferred onto a polyvinylidene fluoride (PVDF) membrane for sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE). The membranes were sealed with 5% nonfat milk containing 0.05% Tween 20 (TBST), incubated with rabbit/mouse polyclonal antibodies against VP2, VP5, VP6, VP7, NS38, NS80, Bcl-2, Caspase-8, Caspase-9 and GADPH, and then, reacted with alkaline phosphatase (AP)-conjugated secondary antibodies. The reaction bands were colored with the AP substrate solution (NBT/BCIP) (Zhang et al., 2022).

2.7. The inhibitory effect of AR on the virus multiplication stage and the virus binding experiment

Initially, CIK cells were inoculated in six-well plates and incubated for 24 h, followed by infection with GCRV (1×10^3 TCID₅₀) for 2 h. The drug was administered 2 h, 4 h, or 6 h after GCRV infection. After incubation for 2 h, the medium containing the drug was absorbed and washed thrice with PBS. Next, cells were added to the cell maintenance solution and incubated for 24 h. The changes in the relative copy number of the virus in various cells were detected by conducting qPCR assays (Hu et al., 2019b). As is known, GCRV can infect cells not only at 28 °C but also at 4 °C (Fang, 2021). Subsequently, we conducted a viral binding experiment to assess whether AR inhibits the proliferation of GCRV by interfering with the adsorption of the virus to the cells. The CIK cells were administered GCRV and AR together and incubated for 30 min at 4 °C and 28 °C, respectively. After incubation, RNA was extracted from these cells to detect the relative copy number of the virus by qPCR.

2.8. Add time analysis

A mixture containing the virus and the drug (80 µg/mL) was incubated at 28 °C for 1, 2, and 4 h, following which, it was added to the cells and incubated for 2 h at 28 °C. After incubation, the supernatant was discarded and the cell maintenance solution containing 2% FBS was added and incubated at 28 °C for 24 h. After the cells were pretreated with the drug for 1, 2, and 4 h, they were infected with the virus for 2 h. The supernatant was discarded, the cell maintenance solution containing 2% FBS was added, and the mixture was incubated at 28 °C for 24 h. In both methods, Trizol reagents were used to extract total RNA, and qPCR was performed to detect changes in the relative copy number of viruses in various cells. The viral titers were determined by plaque assay using CIK cells.

2.9. Observation of cell morphology and nuclear damage

To visualize the damage to the nucleus, the cells were inoculated in a six-well culture plate for 24 h, infected with GCRV (1×10^3 TCID₅₀), and added after 2 h to the drug maintenance solution diluted with 2% FBS MEM (80 µg/mL). After 24 h of culture, the cells were collected, stained with 10 µM Dio for 20 min, fixed with 4% paraformaldehyde (PFA) (servicebio, China), and stained with 1 µg/mL DAPI for 5 min (Song et al., 2020). Dio (3,3'-diiodo-4,4'-dimethoxydiphenylmethane) is usually used as a fluorescent probe for green fluorescent cell membrane labeling. DAPI (4', 6-diamidino-2-phenylindole) is a fluorescence probe that stains the nucleus blue. The cells were washed thrice with 0.1 M PBS after each step, and fluorescence was detected using an inverted fluorescence microscope (SOPTOP ICX41, China).

2.10. Apoptosis assay

Based on the method described in Section 2.9, after 24 h of treatment, cells were harvested and repeatedly washed with PBS buffer. Then, they were analyzed by flow cytometry with the Annexin V/Propyl iodide (PI) staining kit (Solarbio, China), following the manufacturer's instructions. Annexin V and PI were used to stain the cell membrane and nucleus, respectively, to distinguish the different stages of apoptosis. After staining, the fluorescence intensity was determined by flow cytometry (CytoFLEX, China).

2.11. Antiviral activity of AR in vivo

In total, 150 juvenile grass carp were divided into three groups and placed in three aquariums (100 L each). All aquariums were disinfected with potassium permanganate and maintained at 28 ± 0.5 °C. The fish were divided into the PBS control group, the GCRV infection group, and the GCRV+AR treatment group, and the following treatments were performed: (1) In the control group, each grass carp was intraperitoneally injected with 40 µL of PBS and fed continuously for five days. (2) Based on the results of the preliminary experiment, each grass carp in the infected group and the treatment group was injected with 40 µL of a mixture containing equal volumes of GCRV+PBS and GCRV+AR peritoneally, respectively, and fed for five days. All fish were fed commercial feed twice daily (Hao et al., 2023). The liver and intestine were collected on days 1, 3, and 5 for viral gene expression, apoptotic gene expression, anti-inflammatory response assay, and histopathological analysis, as described in Section 2.5. To assess GCRV-induced histopathology, the liver and intestinal tissues of the fish were fixed with 4% PFA on day 5. After fixation, the liver and intestinal tissues were dehydrated with ethanol at different concentration gradients (80%, 85%, 90%, 95%, 100%, 100%) for 30 min each time. After being treated with xylene for 20, 15, and 15 min, the liver and intestine were subjected to paraffin embedding, section, and hematoxylin-eosin (HE) (Servicebio, China) staining. Finally, the slices were sealed, and the liver and intestinal tissue morphology changes were observed and photographed.

2.12. Statistical analyses

All graphs were plotted using GraphPad Prism 8 (GraphPad Software, San Diego, USA). The differences in various parameters between groups were determined by conducting Student's *t*-tests in SPSS v13.0 (SPSS Inc., USA). All data were presented as the mean $\pm$ standard deviation (SD). All results were considered to be statistically significant at $P \leq 0.05$.

3. Results

3.1. Antiviral activity of AR against GCRV infection in CIK cells

The maximum safe concentration of AR on CIK cells was determined

by conducting the CCK-8 assay. The cytotoxicity of AR was negligible at the concentration of 125 µg/mL after 24 h co-incubating, and the cell survival rate was above 90% (Supplementary Fig. 1 A). As shown in Supplementary Fig. 1B, the level of expression of the *S4* gene of GCRV decreased as the concentration of AR increased, suggesting that the antiviral effect of AR is dose-dependent. To elucidate the mechanism underlying the inhibitory effect of AR on GCRV, three treatments were set up. Under an inverted microscope, cytopathic effect (CPE) in CIK cells was detected at 24 h after GCRV infection. No CPE was observed in the control cells. AR inhibited GCRV-induced CPE within 24 h in all three ways. However, the antiviral effect of AR in the G-A group was stronger than that in the GA and A-G groups (Fig. 1A). Consistent results were seen in virus titer measurements. GCRV titers were $10^{3.74}$ (24 h post infection (hpi)) TCID₅₀/0.1 mL. In contrast, GCRV titers were A-G: $10^{1.25}$ (24 hpi), GA: $10^{0.94}$ (24 hpi), and G-A: $10^{0.06}$ (24 hpi) TCID₅₀/0.1 mL in the AR treated group (Fig. 1B). The results of the Western blotting analysis showed that the three treatment methods partially inhibited the expression of VP2, VP5, VP6, VP7, NS38, and NS80 proteins of GCRV at 24 hpi. The expression of viral protein in the G-A group was significantly better than that in the GA and A-G groups (Fig. 1C). These results suggest that AR can significantly inhibit the replication of GCRV in CIK cells.

3.2. AR can affect the infectivity of viruses and can be used to prevent viral invasion

In order to fully elucidate the inhibitory mechanism of AR on GCRV replication, GCRV and AR were incubated extracellular for 1, 2, and 4 h, and then inoculated into CIK cells. After crystal violet staining, the results of CPE showed that a longer co-incubation time of AR and GCRV was associated with less CPE formation by GCRV on CIK cells (Fig. 2A).

The results of qPCR assays showed that the mRNA expression of *S4* of GCRV decreased with an increase in the co-incubation time of AR and GCRV (Fig. 2B), indicating that AR reduces the infectivity of the virus and inhibit the invasion of GCRV. CIK cells were pretreated with AR for 1, 2, and 4 h, then infected with GCRV. As shown in Fig. 2C, the CPE formed by GCRV on CIK cells decreased as the pretreatment duration of the CIK cells with AR increased. At the same time, the level of expression of the *S4* of GCRV decreased with an increase in the AR pretreatment time (Fig. 2D). The above results suggested that the co-incubation of viruses and AR *in vitro* can affect the infectivity of viruses and the pre-incubation of drugs can protect cells from virus infection to a certain extent.

3.3. AR mainly inhibited early replication of GCRV

After finding that AR can effectively inhibit the proliferation of GCRV, AR was added at different stages of the GCRV replication cycle while incubating GCRV-infected cells at 28 °C. As shown in Fig. 3A, cells treated with AR at 0–2 h were almost the same as those not infected by the virus, and the number of CPE was significantly less at 2–4 h than at 4–6 h. The qPCR results showed that administering AR at 0–2 h and 2–4 h after viral infection significantly inhibited the replication of GCRV compared to that in the control group. However, mRNA expression increased sharply when AR was added at 4–6 h compared to its level of expression at 0–2 h and 2–4 h (Fig. 3B). To further confirm the inhibitory effect of AR in virus replication cycle, viral titers were determined. As shown in Fig. 3C, viral titers were significantly increased at 2–4 h and 4–6 h than at 0–2 h. GCRV titers were 100.01 (24 hpi), 100.13 (24 hpi) and 101.01 (24 hpi) TCID₅₀/0.1 mL in the AR treated group at 0–2 h, 2–4 h and 4–6 h, in contrast, GCRV titers were 103.72 (24 hpi) TCID₅₀/

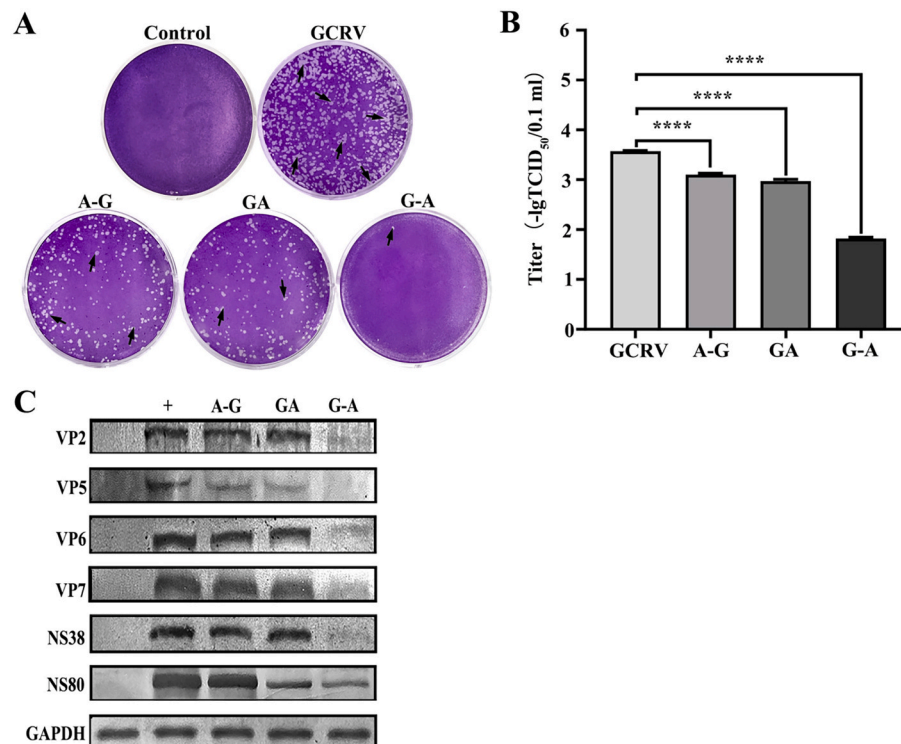


Fig. 1. The same dose of DMSO was administered to the GCRV group and the AR treatment group (0.8%). (A) The inhibitory effect of AR on GCRV in CIK cells. CPE of GCRV on infected CIK cells and morphological protection after treatment with AR. Compared to the normal control group at 24 h, respectively. CPE is shown by a black arrow. A-G: CIK cells were pretreated with AR for 2 h, challenged with GCRV for 2 h, and then, incubated with the cell maintenance medium at 28 °C for 24 h. GA: AR and GCRV were co-incubated at 28 °C for 2 h, and then, added to adherent CIK cells for 2 h. Later, they were replaced with the cell maintenance solution and incubated at 28 °C for 24 h. G-A: The CIK cells were infected with GCRV for 2 h, the drug maintenance solution was replaced, and the cells were incubated at 28 °C for 24 h. (B) AR reduced the titers of GCRV in CIK cells. The data are shown as the Mean ± SD. **: $P < 0.01$ and ****: $P < 0.0001$. (C) The expression of VP2, VP5, VP6, VP7, NS38, and NS80 albumen in GCRV-infected CIK cells treated with AR; GAPDH served as an internal reference albumen.

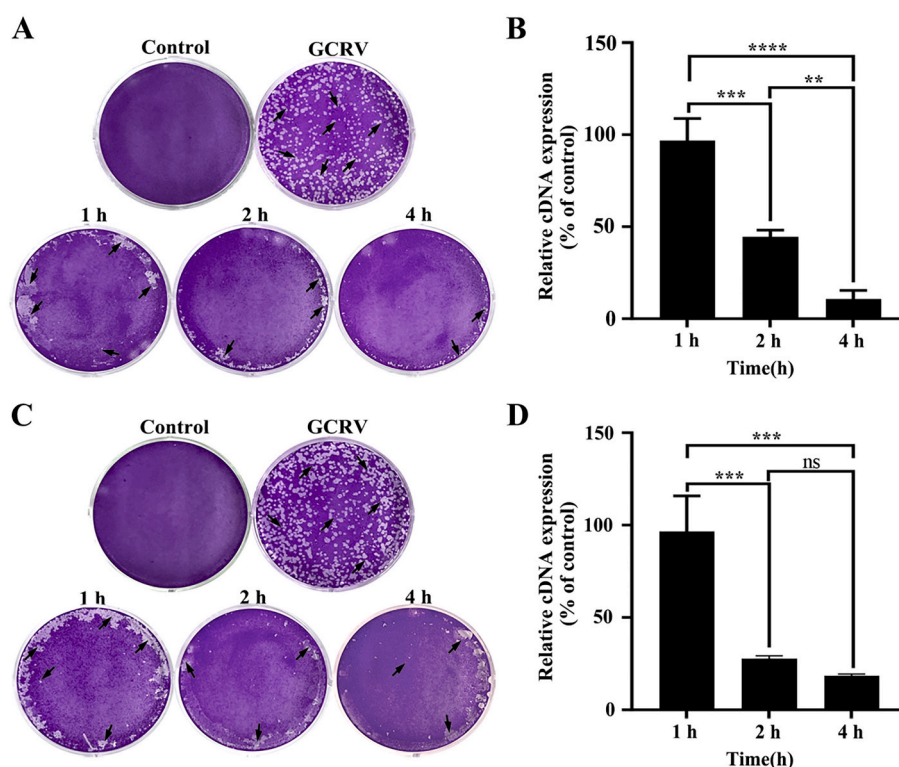


Fig. 2. AR was co-incubated with GCRV. The same dose of DMSO was administered to the GCRV group and the AR treatment group (0.8%). (A) In the co-incubation experiment, AR inhibited the formation of CPE. CPE is shown by a black arrow. Pretreatment GCRV with AR induced antiviral effects on CIK cells. (B) The level of expression of the *S4* gene changed after GCRV was Pre-incubated with AR for 1, 2, and 4 h. (C) CPE was suppressed after CIK cells pretreating with AR. CPE is shown by a black arrow. (D) The level of expression of the GCRV *S4* gene changed in cells after pretreatment with AR for 1, 2, and 4 h. β -actin served as an internal reference gene. All data are shown as the Mean $\pm$ SD; **: $P < 0.01$, ***: $P < 0.001$, and ****: $P < 0.0001$.

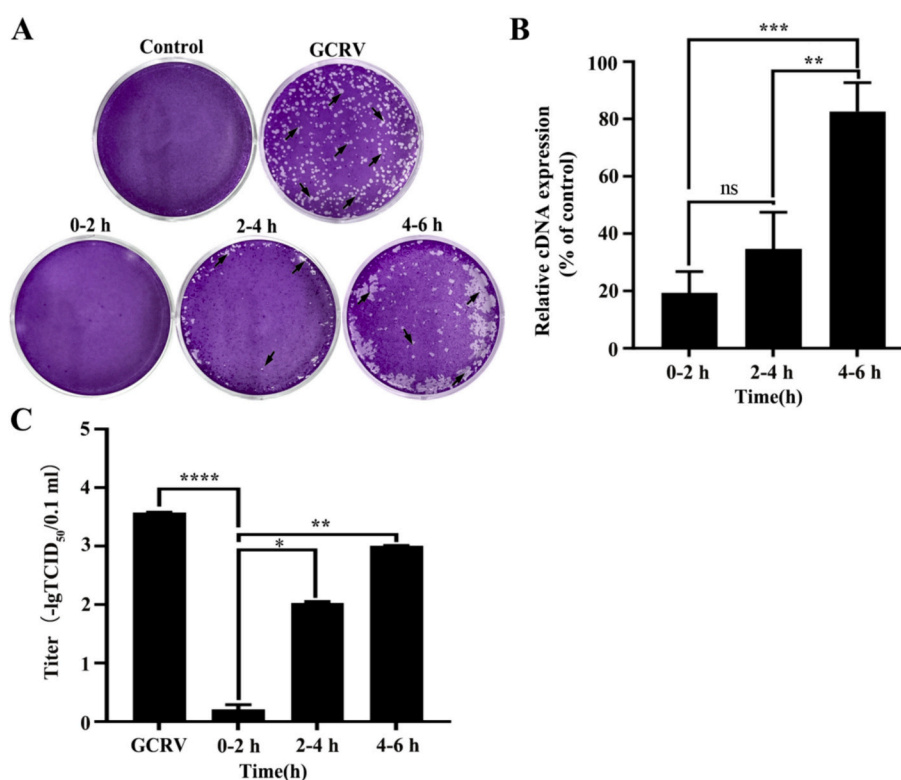


Fig. 3. AR inhibited GCRV infection at an early stage of the viral life cycle. The same dose of DMSO was administered to the GCRV group and the AR treatment group (0.8%). (A) Cell CPE after adding AR maintenance solution at different time points. CPE is shown by a black arrow. (B) The qPCR analysis of the *S4* gene of GCRV. (C) AR reduced the titers of GCRV in CIK cells. All data are shown as the Mean $\pm$ SD; **: $P < 0.01$; ***: $P < 0.001$.

0.1 mL. The results of viral titers were consistent with that of the CPE and the level of expression of the GCRV S4, suggested that AR inhibited the early replication of GCRV.

3.4. AR did not interfere with the adsorption of GCRV

A viral replication cycle can be generally divided into the following processes: adsorption, invasion, hulling, biosynthesis, assembly, and release. In order to further determine whether AR inhibition of early GCRV replication is caused by interference with GCRV adsorption to CIK cells, viral binding experiments were conducted. The results showed that the difference in the expression of the virus between the AR treatment group and the virus group was not significant at 4 °C (Fig. 4A) and 28 °C (Fig. 4B), indicating that AR mainly inhibited the early replication of GCRV but did not interfere with the adsorption of GCRV.

3.5. Effect of AR on apoptosis induced by GCRV

After the virus-infected cells were stained with DAPI and Dio, we assessed different features associated with apoptosis, such as morphological disappearance, nuclear fragmentation, and cytoplasmic degradation. Fig. 5A and B show that although the cells maintained their normal spindle shape, apoptosis was weakened in AR-treated cells. And the difference in the number of the non-apoptotic cells between the treatment groups and the normal blank group was not significant (Fig. 5C). The results of flow cytometry assays showed that AR decreased the apoptosis rate of CIK cells infected with GCRV after 24 h. The apoptosis rate of the CIK cells in the AR-treatment group was reduced by 8.71% compared to the GCRV group, and the difference was highly significant. The apoptosis rate of the CIK cells treated with AR alone was similar to that of normal CIK cells (Fig. 6A and B). These results indicated that AR could protect the integrity of CIK cells from GCRV-induced nuclear fragmentation, cytoplasmic degradation, and apoptosis.

Subsequently, the expressions of apoptosis-related genes were quantitatively detected by fluorescence. AR inhibited the expression levels of pro-apoptotic genes. Compared to the uninfected cells, the expression levels of Caspase-3, Caspase-8, Caspase-9, and Bax in the GCRV group increased 11.55-fold, 1.35-fold, 19.03-fold, and 7.95-fold, respectively. At the same time, the expression level of Bcl-2 decreased by 0.4-fold. On the contrary, the expression level of Bcl-2 was significantly upregulated by 1.7-fold in the G-A group, while the expression

level of Bax was down-regulated considerably by 2.21-fold. In addition, the expression levels of Caspase-3, Caspase-8, and Caspase-9 were restored by AR. In the presence of AR, the expression levels of Caspase-3, Caspase-8 and Caspase-9 was reduced by 5.69-fold, 0.46-fold and 11.32-fold respectively (Fig. 7A). Western blotting experiments confirmed that AR significantly inhibited the expression of pro-apoptotic protein Caspase-8 and Caspase-9, and enhanced the expression of anti-apoptotic protein Bcl-2 in the G-A group, which was similar to the fluorescence quantitative results (Fig. 7B), suggesting that AR inhibits GCRV replication mainly by blocking GCRV-induced apoptosis.

3.6. Antiviral effects of AR on grass carp infected with GCRV

To assess the effect of AR in GCRV infection *in vivo*, grass carp were treated with AR, and the viral loads in the liver and intestine of grass carp were measured one, three, and five days after GCRV infection. The viral load in the liver and intestine of the fish in the drug treatment group was higher than that in the GCRV group at 1 day postinfection (dpi), which may be related to the body's internal environment. The viral loads in the liver and intestine of G + A group were significantly lower than those in the GCRV-infected group at 3 and 5 dpi (Fig. 8A and B). As shown in Fig. 8C, histopathological lesions (inflammation in the liver, atrophy and rupture in the intestinal villi) induced by GCRV were observed in the GCRV-infected grass carp. In the AR-treated group, the structural integrity of the liver was restored to normal values and inflammation-related symptoms improved significantly. The structure of the intestinal villi was restored considerably in the AR-treated group, compared to the corresponding structure in the GCRV-infected group. However, the structural integrity was not as good as that of the control group. These results indicated that AR had therapeutic effects on grass carp infected with GCRV, and may protect grass carp tissues and organs from the virus.

3.7. AR regulates the expression of apoptosis genes *in vivo*

It was confirmed that AR mainly achieves antiviral effects by inhibiting GCRV-induced apoptosis *in vitro*. We also evaluated the expression of apoptosis genes by AR *in vivo*. The results showed that in the liver, the AR treatment group could effectively reduce the expressions of pro-apoptotic genes Caspase-3, Caspase-8, Caspase-9 and Bax, and up-regulate the Bcl-2 gene (Fig. 9A). In the gut, AR treatment group also significantly inhibited the expression of pro-apoptotic genes and upregulated Bcl-2 compared to the GCRV group (Fig. 9B). Consistent with the results of cell experiments *in vitro*, AR can also inhibit GCRV-induced apoptosis *in vivo*.

3.8. AR effectively improved inflammation in grass carp

After finding the role of AR in anti-inflammatory action *in vivo*, it is essential to measure the relevant pro-inflammatory factors further to determine the inhibitory effect of AR on inflammation. As shown in (Fig. 10A), the expression of the IL-1 β gene in the liver of AR-treated grass carp was downregulated by 3.32, 1.9, and 2.2-fold compared to its expression in the GCRV-infected grass carp on 1, 3 and 5 dpi. And the expressions of IL-1 β gene in the intestinal tract of the G + A group by 4.22, 13.1, and 6.6-fold, respectively (Fig. 10B). The expression of the IL-6 gene decreased significantly in the liver of the G + A group on 1 and 3 dpi (Fig. 10C), and in the intestine on 3 and 5 dpi (Fig. 10D). The level of TNF- α was also significantly downregulated in the liver and intestine, with 1.43, 1.81, and 2.568-fold downregulated in the liver (Fig. 10E), and 1.52, 1.93, and 1.58-fold downregulated in the intestine on 1, 3, and 5 dpi, respectively (Fig. 10F). All together, the results indicated that AR can effectively downregulated pro-inflammatory factors and impart protection against viruses to the body while reducing the occurrence of inflammation.

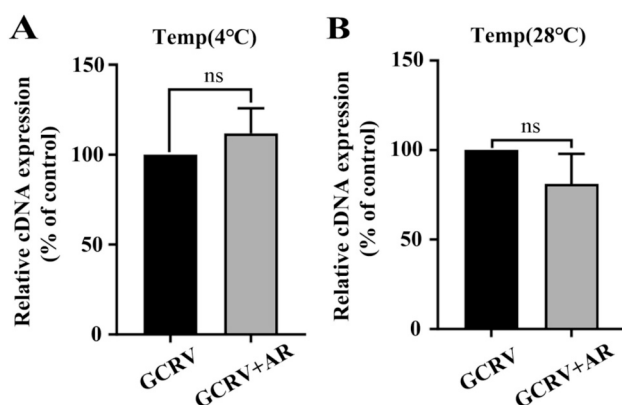


Fig. 4. The adhesion of GCRV to cells was not affected by AR. CIK cells were infected with a mixture of GCRV and AR at 4 °C and 28 °C for 30 min. The virus adhered to the cells but could not enter and replicate in the cells. After 30 min, the cells were cleaned and harvested, and qPCR analysis of the S4 gene was performed to quantify the viruses that adhered to the cells. The same dose of DMSO was administered to the GCRV group and the AR treatment group (0.8%). The S4 gene was assessed by qPCR analysis (A) at 4 °C, and (B) at 28 °C; β -actin served as an internal reference gene. All data are shown as the Mean $\pm$ SD.

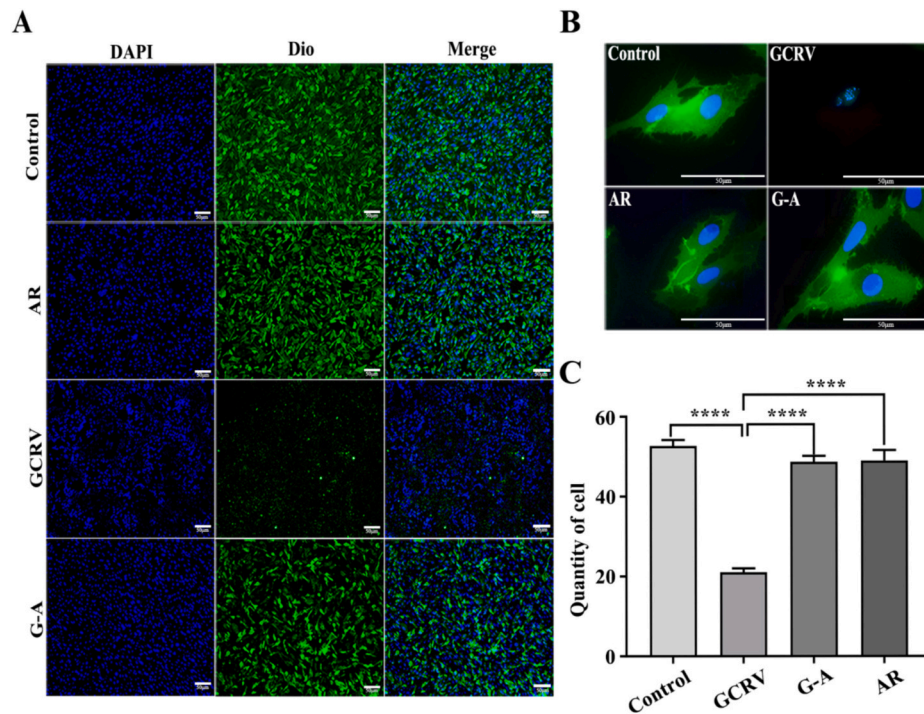


Fig. 5. Changes in the nucleus and cytoplasm after 24 h. The same dose of DMSO was administered to the GCRV group and the AR treatment group (0.8%). (A and B) The cells were co-stained with Dio and DAPI; scale bar: 50 μ m. The experiment was repeated thrice with similar results. (C) Quantity of cells. The data are shown as the Mean $\pm$ SD, **: $P < 0.01$ and ****: $P < 0.0001$.

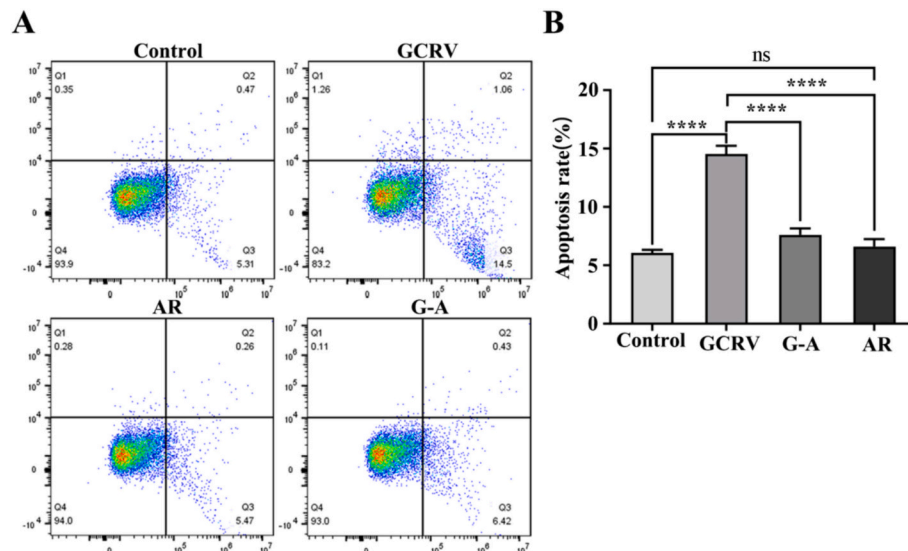


Fig. 6. The same dose of DMSO was administered to the GCRV group and the AR treatment group (0.8%). (A) AR inhibited apoptosis induced by GCRV in CIK cells. The percentage of cells that were viable, or in the early apoptosis and late apoptosis/necrosis stages were assessed by Annexin V/PI staining 24 h after infection. Flow cytometry analysis was performed to detect apoptosis after the CIK cells were stained with Annexin V/PI. The experiment was repeated thrice, and similar results were recorded. (B) Apoptosis ratio. All data are shown as the Mean $\pm$ SD; **: $P < 0.01$ and ****: $P < 0.0001$.

4. Discussion

Aquatic viruses can seriously threaten the aquaculture industry by causing the large-scale death of aquatic animals. Grass carp is a popular freshwater fish and a highly promising aquaculture species. However, they suffer from serious bleeding symptoms after getting infected with GCRV, which leads to the large-scale death of fish fry (Wang et al., 2013). This virus is the most pathogenic AqRV and has caused substantial economic losses to the breeding industry (Hongli et al., 2014;

Rangel et al., 1999). Many research reports have investigated ways to develop vaccines against GCRV (Samanta et al., 2013). Four vaccine types have been studied to control GCRV, including inactivated live attenuated, recombinant subunit, and DNA (Chen et al., 2018; Pei et al., 2019). However, inactivated virus cell culture vaccines have replaced tissue-culture vaccines due to their inconsistent quality, and the risk of incomplete inactivation and poor immune protection in some cases. The recombinant subunit vaccine is an improved version of the inactivated virus vaccine. Live attenuated vaccines and DNA vaccines are effective

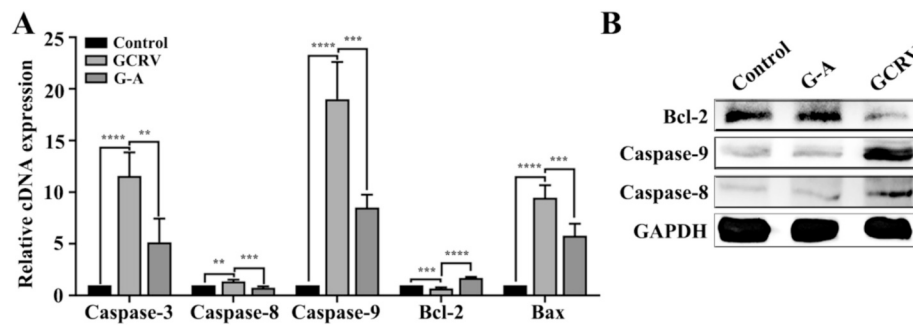


Fig. 7. The same dose of DMSO was administered to the GCRV group and the AR treatment group (0.8%). (A) Expression levels of Caspase-3, Caspase-8, Caspase-9, Bcl-2 and Bax genes. β -actin served as an internal reference gene. The data are shown as the Mean $\pm$ SD. **: $P < 0.01$ and ****: $P < 0.0001$. (B) Expression levels of Caspase-8, Caspase-9 and Bcl-2 proteins. GAPDH served as an internal reference albumen.

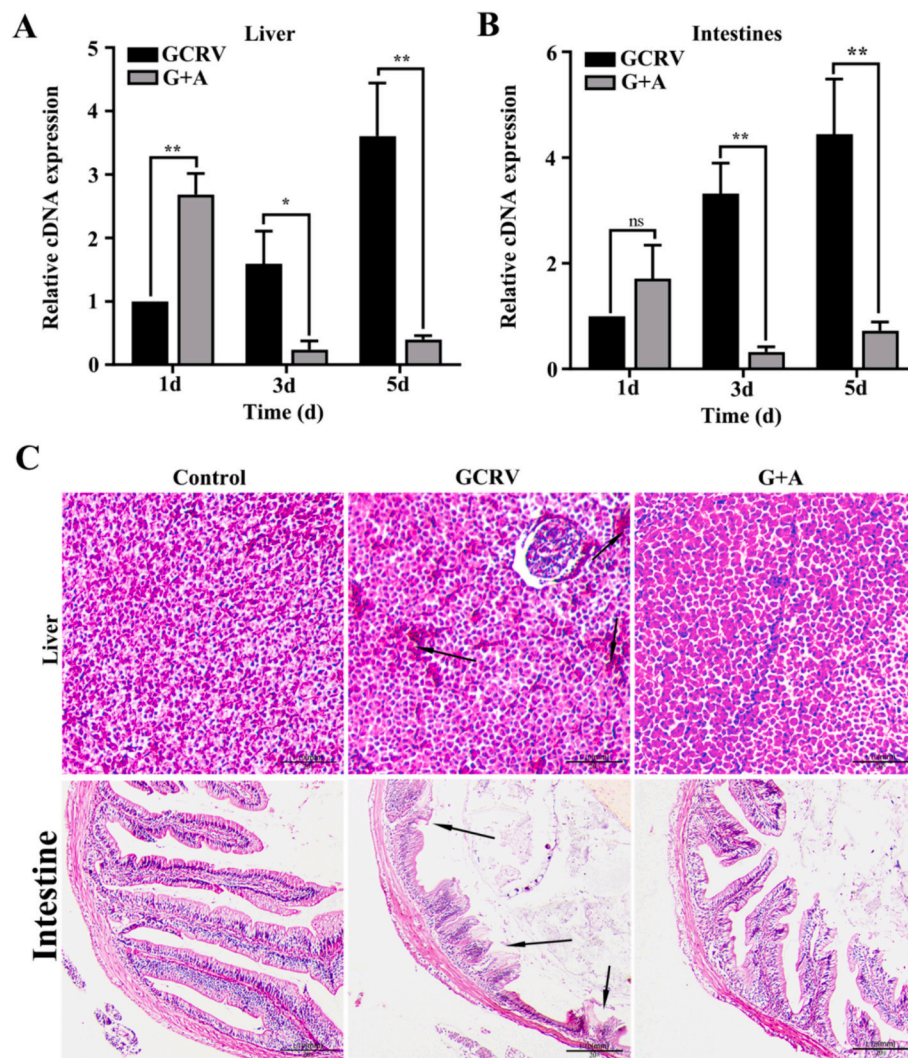


Fig. 8. Antiviral activity of AR *in vivo*. The same dose of DMSO was administered to the GCRV group and the AR treatment group. (A) A viral load analysis was performed on the liver of grass carp on days 1, 3, and 5. (B) A viral load analysis was performed on the intestine of grass carp on days 1, 3, and 5. β -actin served as an internal reference gene. All data are shown as the Mean $\pm$ SD; *: $P < 0.05$ and **: $P < 0.01$. (C) The effect of intraperitoneal injection of AR on histopathological lesions induced by GCRV. Histopathological damage induced by GCRV in the liver and intestine on day 5. The black arrows indicate inflammation in the liver and intestinal villi atrophy and damage.

against fish viruses. However, because of significant differences between strains, some vaccines are rendered weak and sometimes impart no protection. When the virus causes severe bleeding after infection, mortality cannot be reduced by vaccination and immune regulation

(Citarasu, 2010). Therefore, effective anti-GCRV drugs need to be developed to compensate for the limitations of vaccines.

Chinese medicine has long been used to treat diseases. The development of natural plant compounds can adequately prevent and control

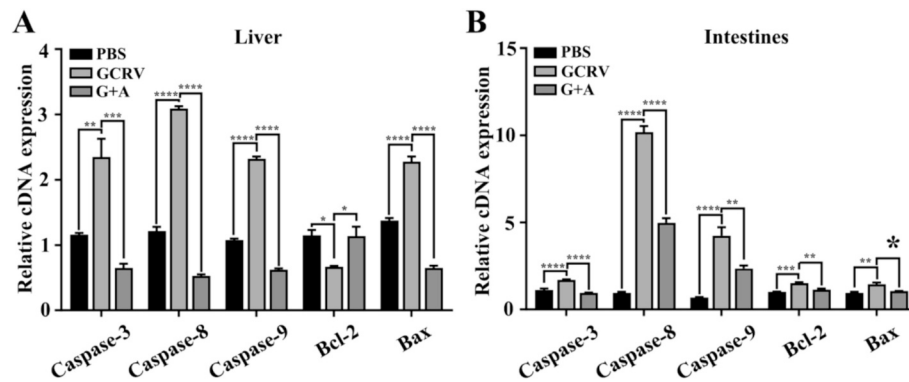


Fig. 9. Expression of apoptotic genes. The same dose of DMSO was administered to the GCRV control group and the AR treatment group. (A) Apoptotic gene expression in the liver. (B) Apoptotic gene expression in intestines. β -actin served as an internal reference gene. The data are shown as the Mean $\pm$ SD; *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, and ****: $P < 0.0001$.

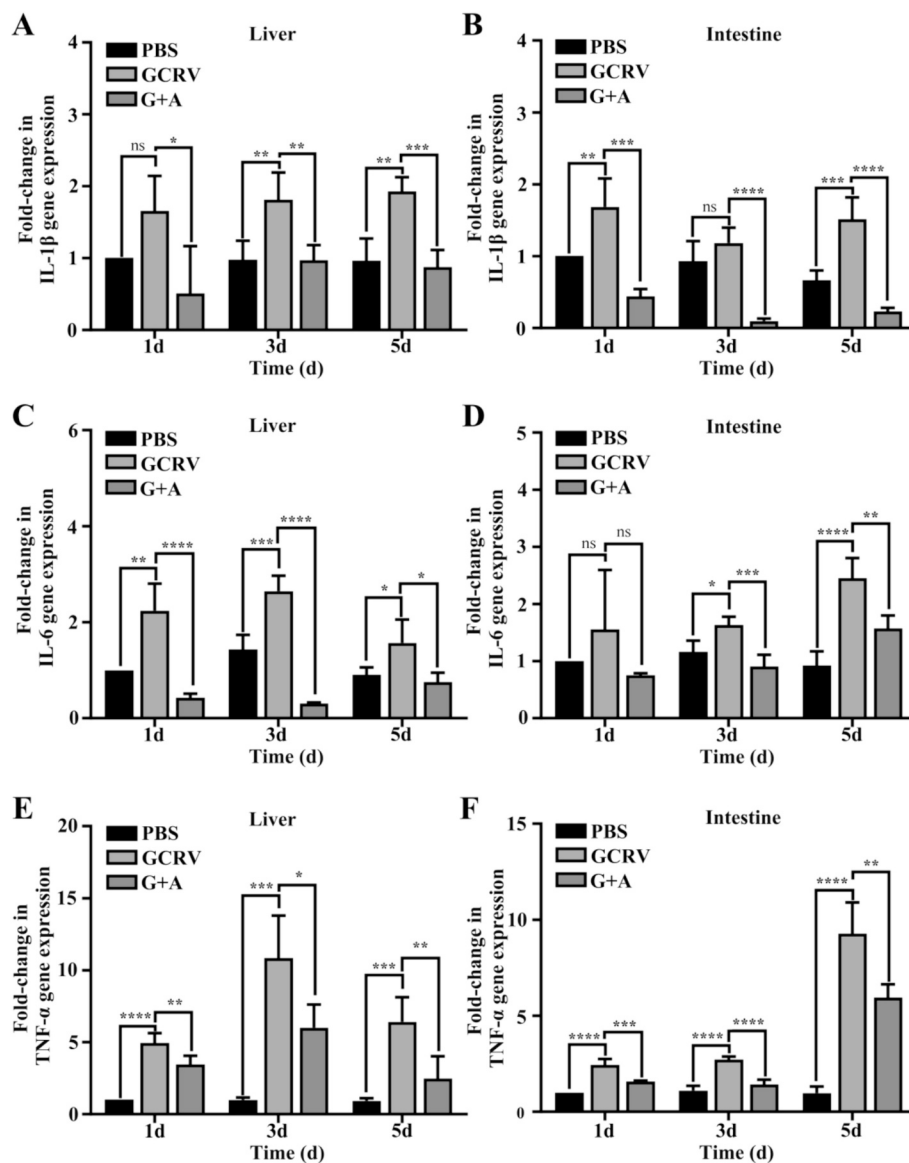


Fig. 10. AR effectively improved inflammation in grass carp. The same dose of DMSO was administered to the GCRV control group and the AR treatment group. (A–B) The expression of IL-1 β in the liver and intestine was assessed after one, three, and five days of intracavitary injection therapy. (C–D) The expression of IL-6 in the liver and intestine was assessed after one, three, and five days of intracavitary injection therapy. (E–F) The expression of TNF- α in the liver and intestine was assessed after one, three, and five days of intracavitary injection therapy. β -actin served as an internal reference gene. The data are shown as the Mean $\pm$ SD; *: $P < 0.05$, **: $P < 0.01$, ***: $P < 0.001$, and ****: $P < 0.0001$.

aquaculture viruses (Zhu, 2020). Epigallocatechin gallate (EGCG), the main theine in green tea, can inhibit viral replication by blocking viral adhesion to CIK cells (Wang et al., 2016). To ensure food security and reduce environmental pollution, researchers should be encouraged to conduct rigorous studies in this area, especially on Chinese herbal compounds, as they are eco-friendly. AR is one of the main active ingredients in burdock seeds. The active ingredients in burdock seeds have been investigated in the past decade because of their strong therapeutic effects.

Since viruses lack enzymes and a suitable system for most metabolic and biosynthetic reactions, thus, they rely on cells for most functions. Some of the molecules involved in specific phases of the viral replication cycle in the cell may be used as targets for antiviral drugs. As GCRV and CIK have been used as *in vitro* infection models, they were used to evaluate the antiviral activity of AR against GCRV and elucidated the mechanism underlying the inhibitory effects of AR on infection in this study. The *S4* gene plays a central role in viral factory-like structures (VFLS) formation and recruitment of the viral component in GCRV particle assembly (Cai et al., 2011). In this study, AR showed strong anti-GCRV effects *in vitro*. GCRV enters the host cell during viral replication through receptor-mediated endocytosis (Zhang et al., 2018b). After adsorption, the virus penetrates the cell membrane using virus-encoded membrane penetrant proteins (Wang et al., 2018). When the virus infects the host, the outer shell, where endogenous transcription occurs *in vivo*, is stripped, and the undercoat remains intact in the cell. The new mRNA released from the inner shell serves as a template for protein synthesis. The newly synthesized structural proteins are recruited from the daughter virus into the viral inclusion body. The proteins constituting GCRV mainly include outer shell proteins (VP5 and VP7), undercoat shell proteins (VP1, VP2, VP3, VP4, and VP6), and non-structural proteins (NS38 and NS80). Among these, five core proteins initiate endogenous transcription and are responsible for viral transcription and replication. Coat proteins directly participate in the interaction between viruses and host cells, whereas, non-structural proteins participate in the viral replication cycle (Attoui et al., 2002; Cheng et al., 2008; Cheng et al., 2010; Fang et al., 2000; Fang et al., 2005; Yan et al., 2014; Yu et al., 2019). This study found that AR could significantly inhibit the expression of viral proteins VP2, VP5, VP6, VP7, NS38 and NS80. By adding time determination *in vitro*, AR can not only effectively reduce the infectivity of the virus, but also protect cells against the invasion of the virus. It mainly affects the early replication of the virus. However, our findings were different from those of previous studies, epigallocatechin gallate (EGCG) can inhibit the adhesion of the particles of the GCRV-I strain to the surface of CIK cells, whereas AR used in this study without affecting the adsorption of the virus. The virus-induced pathogenesis is related to oxidative damage and apoptosis (Jia et al., 2014; Valyi-Nagy and Dermody, 2005). Death receptor and mitochondrial pathway are the two main pathways that activate Caspase. The death receptor junction initiates the death receptor pathway and then recruits the junction molecule to process the downstream Caspase to stimulate its own Caspase precursor. The activation of mitochondrial pathways mainly depends on organelle dysfunction, and Bcl-2 family proteins play a major role in regulating mitochondrial dysfunction (Dempsey et al., 2003). In response to environmental cues, Bax family proteins mainly regulate the permeability of mitochondrial outer membrane, releasing cytochrome *C* from mitochondria (Danial and Korsmeyer, 2004). Cysteine protease activation is mainly carried out in a stratified form, such as the activation of Caspase-3, which leads to apoptosis. Cells infected by GCRV also undergo apoptosis (Chu et al., 2020; Dai et al., 2022; Liang and Su, 2019). The apoptosis of host cells facilitates the release of assembled virus particles (Li et al., 2019). Studies have shown that GCRV infection induces apoptosis in early CIK cells by activating promoters Caspase-8, caspase-9 and effector caspase-3, which proves that apoptosis depends on the activation of endogenous and exogenous pathways of Caspase (Jia et al., 2014). We found that AR-treated cells infected with GCRV, protected cell morphology, preserved

the normal spindle appearance of cells, and prevented most of the nuclear damage caused by GCRV in CIK cells. After GCRV infection, both death receptors and mitochondrial pathways are triggered, and many apoptotic signals are released. However, it is noteworthy that GCRV-induced apoptosis and cell damage were not observed in AR-treated CIK cells. AR has a good intervention effect on the process of GCRV infection of CIK by up-regulating Bcl-2 gene expression and down-regulating Bax gene expression and protein expression, suggesting that AR can affect GCRV's regulation of Bcl-2 family protein-induced apoptosis. In addition, AR also inhibited the gene expression and protein expression of Caspase-3, caspase-8, and Caspase-9 in GCRV-induced CIK cells. Previous studies have shown that coumarin compounds can reduce apoptosis and virus-activated caspase-3, 8, and 9 activity in SVCV-infected cells (Liu et al., 2017; Shen et al., 2018). In conclusion, our results prove that the activation of Caspase in the death receptor pathway and mitochondrial pathway are related to GCRV infection. However, AR not only has a certain protective effect on CIK cells but also inhibits Caspase activity. These findings suggested that AR exhibits antiviral effects by blocking GCRV-induced apoptosis and limiting viral release.

In aquaculture, AR is administered in three main ways: oral administration, immersion, and injection. A suitable method of administration needs to be determined for the drugs to exert adequate levels of antiviral activity. Sick aquatic organisms are commonly treated with herbs or herbal extracts by incorporating them into the diet, making it an inexpensive strategy (Elumalai et al., 2020). However, as many sick aquatic animals do not eat, new routes of administration need to be determined. Although immersion is simpler than oral administration, it requires larger quantities of drugs, works slowly, and may also promote microbial resistance (Fu et al., 2007; Quesada et al., 2013). In contrast, injecting drugs might be the most effective method (Terzi et al., 2020). We found that the intraperitoneal injection of AR reduced the viral load of GCRV-infected grass carp, which suggested that AR has antiviral activity *in vivo*. The source of GCRV infection is another infected grass carp, and the main route of infection is horizontal transmission, or vertical transmission through eggs. Coumarin derivatives B4 and C3007 can reduce mortality and viral load in spring viremia of carp virus (SVCV) infected fish, thus limiting horizontal transmission to some extent (Liu et al., 2020; Qiu et al., 2020). Although AR cannot completely inactivate the virus, it may reduce horizontal transmission. Our findings suggested that the application of AR might be considered a preventative measure against virus transmission in aquaculture settings.

The innate immunity of aquatic animals protects the host against pathogen invasion. The pattern recognition receptors (PRRs) in fish are evolutionarily conserved and responsible for sensing the presence of pathogen-associated molecular patterns (PAMPs) (Aoki et al., 2013; Takeuchi and Akira, 2010). Inflammation is a highly regulated defense process that involves the release of cytokines, chemokines, and growth factors, as well as migrating inflammatory cells, such as neutrophils, monocytes, and lymphocytes from the blood to the affected tissue. We found that GCRV infection can induce inflammation, and AR can effectively alleviate the inflammation caused by GCRV in grass carp. We also examined the expression of different classes of cytokine genes, including TNF- α , IL-1 β , and IL-6. All three are potent pro-inflammatory factors that regulate other molecules and inflammation in the early stages of viral infection (Secombes et al., 2001). Some researchers have shown the anti-inflammatory activity of Arctigenin (Hyam et al., 2013; Song et al., 2016). We also found that AR can strongly inhibit the expression of GCRV-induced pro-inflammatory factors. Our results suggested that AR can induce the expression of virus-related genes and strengthen the host immune response.

5. Conclusions

AR can weaken the cell CPE and apoptosis induced by GCRV, reduce the viral load of grass carp, and induce antiviral effects against GCRV *in*

vitro and *in vivo*. AR effectively inhibited the proliferation and protein expression of GCRV *in vitro*. AR can protect cells, reduce virus infectivity, prevent virus invasion, and block early viral replication, but it does not interfere with viral adsorption. AR inhibits GCRV replication mainly by blocking GCRV-induced apoptosis *in vitro*. AR can also achieve antiviral effects by regulating related apoptosis factors *in vivo*. In addition, AR can also reduce the inflammatory response caused by GCRV infection by activating the innate immune function of the host. Our findings suggest that AR can be used as a therapy for aquatic systems and as an immune booster to prevent virus transmission in aquaculture settings. These antiviral mechanisms reveal new ways to implement AR as an anti-GCRV agent in aquaculture.

Ethics statement

All animal-involving experiments of this study were approved by the Animal Care and Use Committee of College of Animal Sciences, Yangtze University, and all efforts were made to minimize suffering.

CRediT authorship contribution statement

Mengran Xu: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Huaming Li:** Writing – original draft, Software, Methodology, Data curation. **Xiaomin Wu:** Visualization, Software, Methodology. **Lu Dai:** Software, Methodology, Data curation. **Rongrong Yang:** Visualization, Software, Formal analysis. **Qiwen Zhang:** Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Zhe Zhang:** Software, Methodology, Formal analysis, Data curation. **Liancheng Lei:** Writing – original draft, Methodology, Conceptualization. **Fuxian Zhang:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aquaculture.2024.741355>.

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